HOSPITAL MANAGEMENT SYSTEM

MySQL Database Implementation – Practical SQL Screenshots

Project database: hospitalmanagementsystem

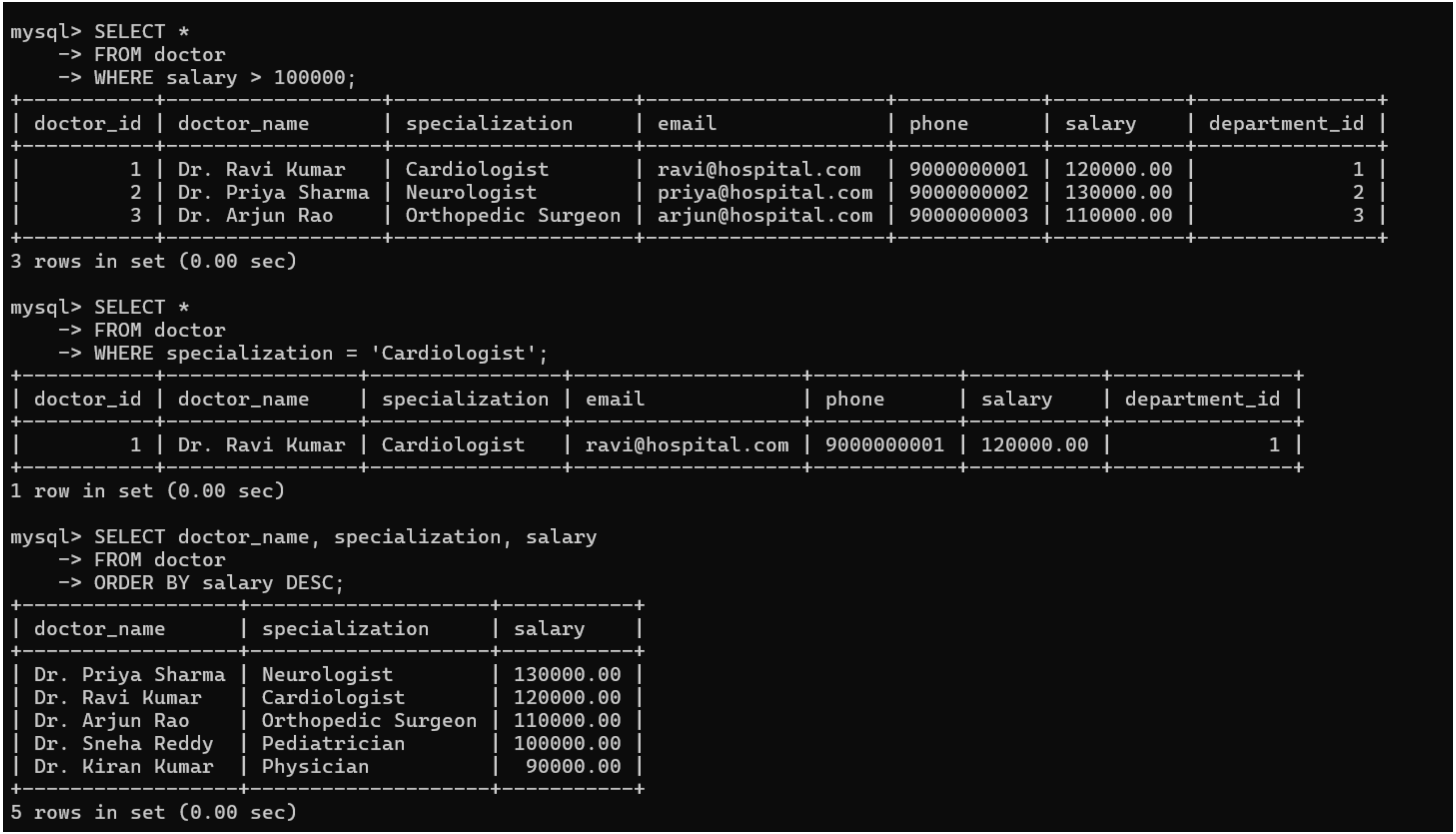
Tables demonstrated: doctor, department and room

SQL concepts: filtering, sorting, aggregate functions, grouping, pattern matching, joins and table constraints

Prepared as a visual record of practical work completed in the MySQL Command Line Client.

1. Filtering and Sorting – WHERE and ORDER BY

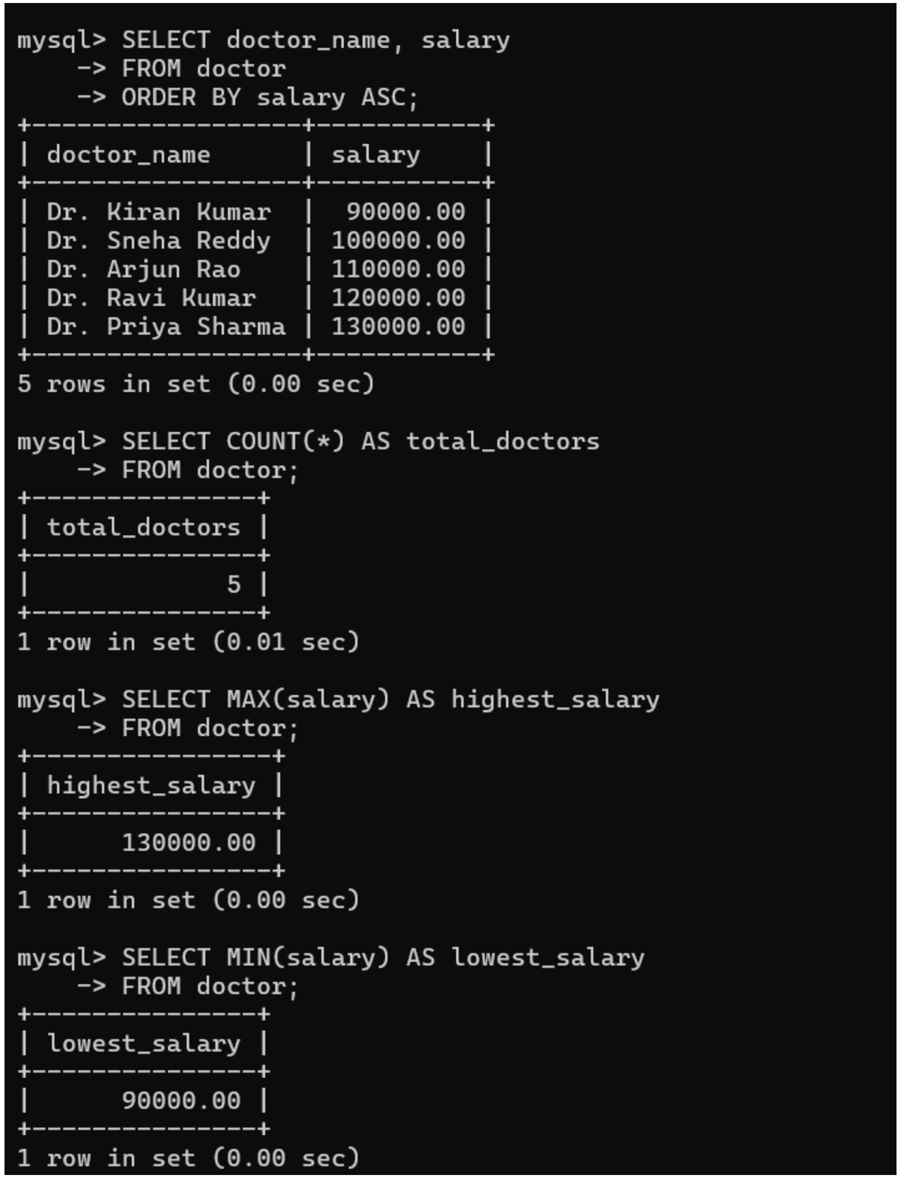
Shows salary filtering, specialization filtering, and ascending/descending salary sorting.



```
mysql> SELECT *
    -> FROM doctor
    -> WHERE salary > 100000;
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
| doctor_id | doctor_name      | specialization     | email              | phone      | salary    | department_id |
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
|         1 | Dr. Ravi Kumar   | Cardiologist       | ravi@hospital.com  | 9000000001 | 120000.00 |             1 |
|         2 | Dr. Priya Sharma | Neurologist        | priya@hospital.com | 9000000002 | 130000.00 |             2 |
|         3 | Dr. Arjun Rao    | Orthopedic Surgeon | arjun@hospital.com | 9000000003 | 110000.00 |             3 |
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
3 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM doctor
    -> WHERE specialization = 'Cardiologist';
+-----------+----------------+----------------+-------------------+------------+-----------+---------------+
| doctor_id | doctor_name    | specialization | email             | phone      | salary    | department_id |
+-----------+----------------+----------------+-------------------+------------+-----------+---------------+
|         1 | Dr. Ravi Kumar | Cardiologist   | ravi@hospital.com | 9000000001 | 120000.00 |             1 |
+-----------+----------------+----------------+-------------------+------------+-----------+---------------+
1 row in set (0.00 sec)

mysql> SELECT doctor_name, specialization, salary
    -> FROM doctor
    -> ORDER BY salary DESC;
+------------------+--------------------+-----------+
| doctor_name      | specialization     | salary    |
+------------------+--------------------+-----------+
| Dr. Priya Sharma | Neurologist        | 130000.00 |
| Dr. Ravi Kumar   | Cardiologist       | 120000.00 |
| Dr. Arjun Rao    | Orthopedic Surgeon | 110000.00 |
| Dr. Sneha Reddy  | Pediatrician       | 100000.00 |
| Dr. Kiran Kumar  | Physician          |  90000.00 |
+------------------+--------------------+-----------+
5 rows in set (0.00 sec)
```

2. Aggregate Functions – COUNT, MAX and MIN

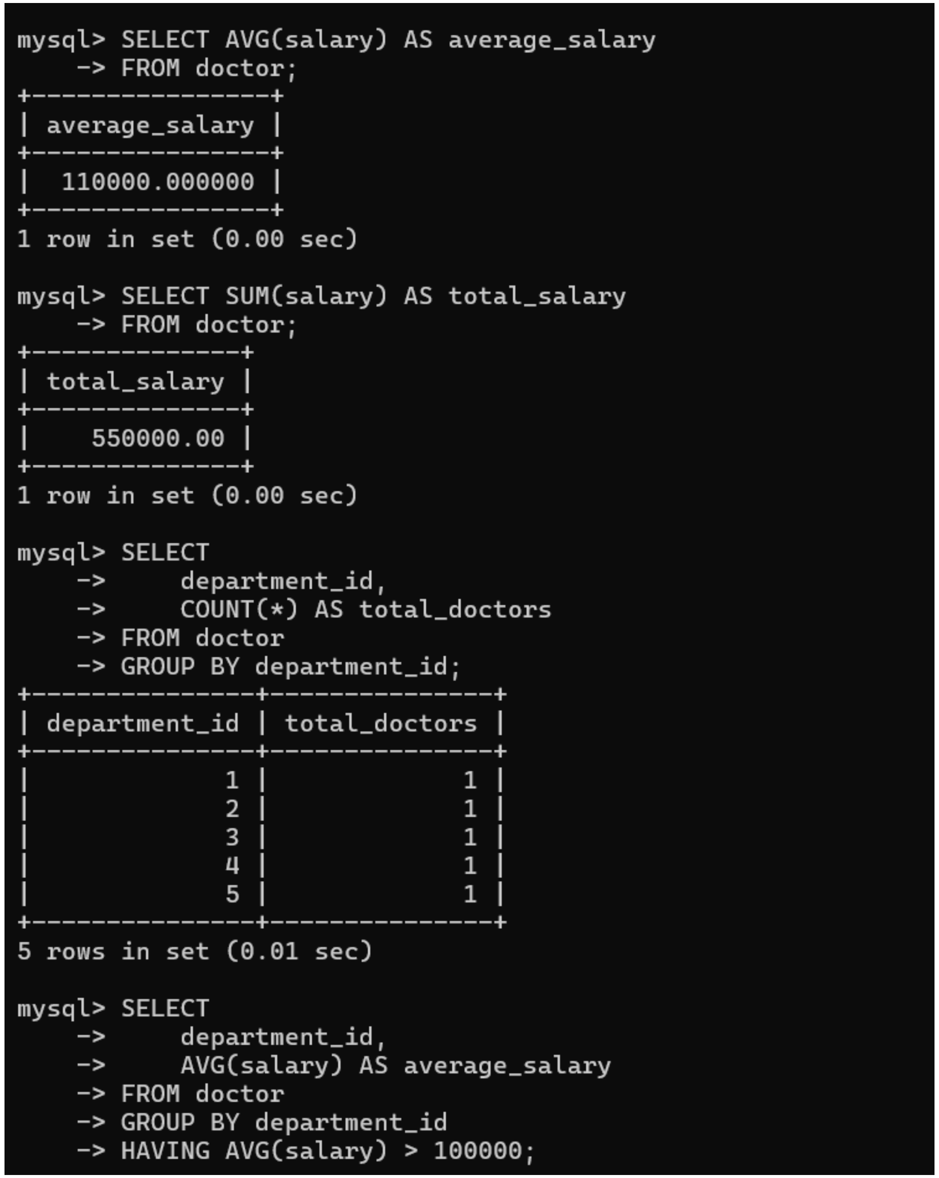
Shows the total number of doctors and the highest and lowest salary values.



```
mysql> SELECT doctor_name, salary
    -> FROM doctor
    -> ORDER BY salary ASC;
+------------------+-----------+
| doctor_name      | salary    |
+------------------+-----------+
| Dr. Kiran Kumar  |  90000.00 |
| Dr. Sneha Reddy  | 100000.00 |
| Dr. Arjun Rao    | 110000.00 |
| Dr. Ravi Kumar   | 120000.00 |
| Dr. Priya Sharma | 130000.00 |
+------------------+-----------+
5 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS total_doctors
    -> FROM doctor;
+---------------+
| total_doctors |
+---------------+
|             5 |
+---------------+
1 row in set (0.01 sec)

mysql> SELECT MAX(salary) AS highest_salary
    -> FROM doctor;
+----------------+
| highest_salary |
+----------------+
|      130000.00 |
+----------------+
1 row in set (0.00 sec)

mysql> SELECT MIN(salary) AS lowest_salary
    -> FROM doctor;
+---------------+
| lowest_salary |
+---------------+
|      90000.00 |
+---------------+
1 row in set (0.00 sec)
```

3. Aggregate Functions and Grouping – AVG, SUM, GROUP BY and HAVING

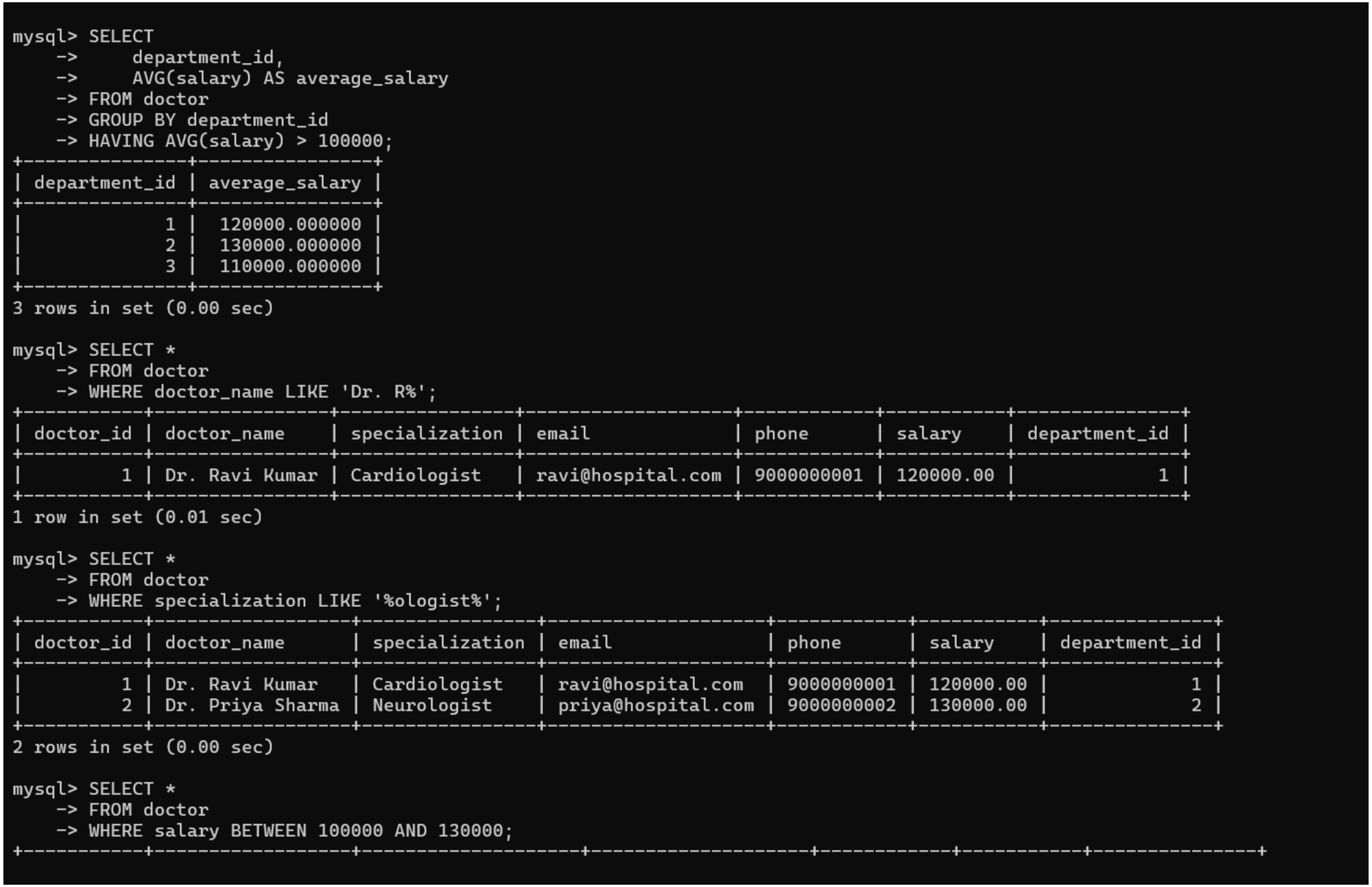
Shows average and total salary calculations, doctor counts by department, and filtering grouped results.



```
mysql> SELECT AVG(salary) AS average_salary
    -> FROM doctor;
+----------------+
| average_salary |
+----------------+
|  110000.000000 |
+----------------+
1 row in set (0.00 sec)

mysql> SELECT SUM(salary) AS total_salary
    -> FROM doctor;
+--------------+
| total_salary |
+--------------+
|    550000.00 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT
    ->     department_id,
    ->     COUNT(*) AS total_doctors
    -> FROM doctor
    -> GROUP BY department_id;
+---------------+---------------+
| department_id | total_doctors |
+---------------+---------------+
|             1 |             1 |
|             2 |             1 |
|             3 |             1 |
|             4 |             1 |
|             5 |             1 |
+---------------+---------------+
5 rows in set (0.01 sec)

mysql> SELECT
    ->     department_id,
    ->     AVG(salary) AS average_salary
    -> FROM doctor
    -> GROUP BY department_id
    -> HAVING AVG(salary) > 100000;
```

4. Pattern Matching and Range Filtering – LIKE and BETWEEN

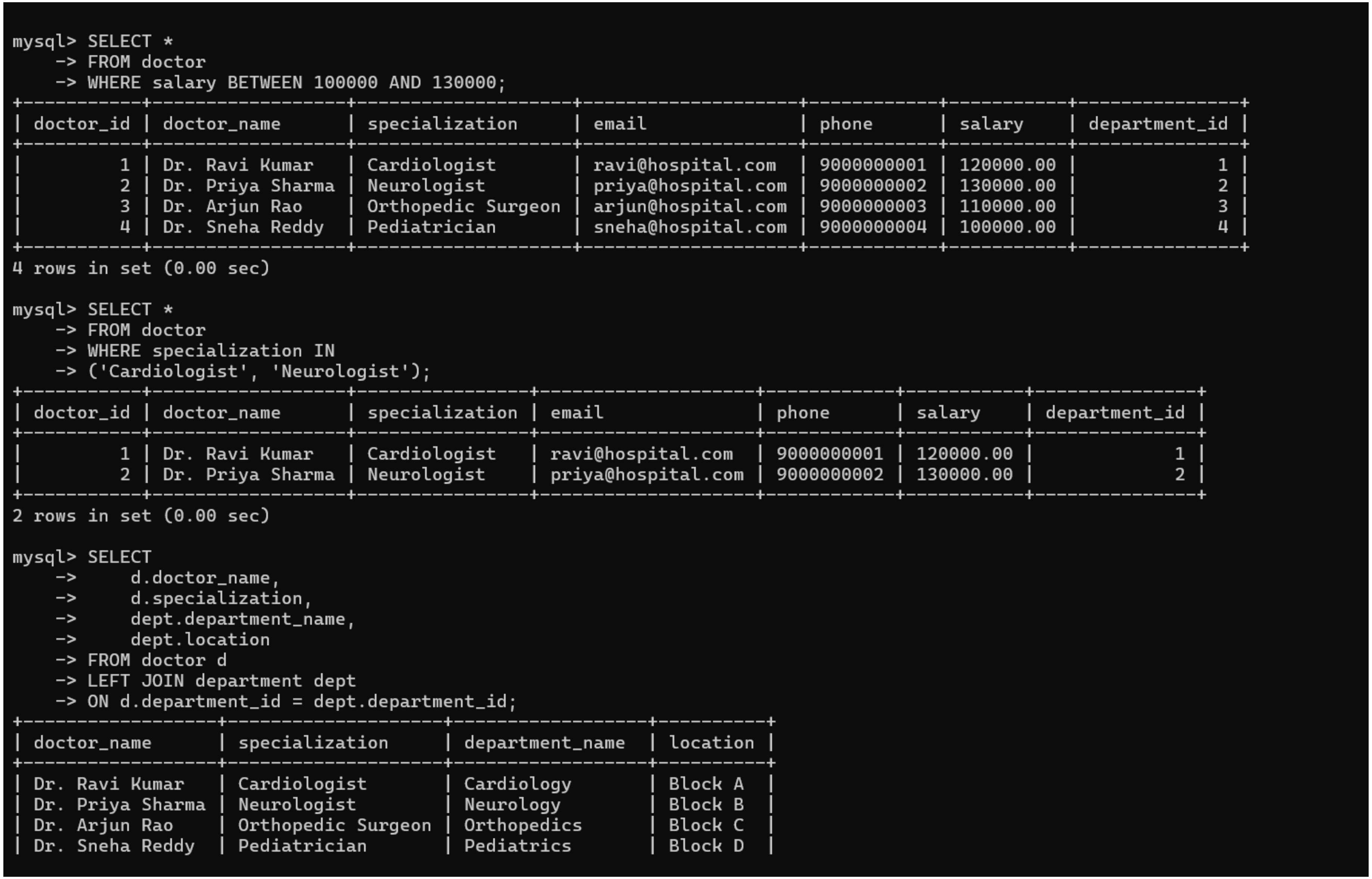
Shows searches using partial text and a salary range. The screenshot also includes the completed LIKE results.



```
mysql> SELECT
    ->     department_id,
    ->     AVG(salary) AS average_salary
    -> FROM doctor
    -> GROUP BY department_id
    -> HAVING AVG(salary) > 100000;
+---------------+----------------+
| department_id | average_salary |
+---------------+----------------+
|             1 |  120000.000000 |
|             2 |  130000.000000 |
|             3 |  110000.000000 |
+---------------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM doctor
    -> WHERE doctor_name LIKE 'Dr. R%';
+-----------+----------------+----------------+-------------------+------------+-----------+---------------+
| doctor_id | doctor_name    | specialization | email             | phone      | salary    | department_id |
+-----------+----------------+----------------+-------------------+------------+-----------+---------------+
|         1 | Dr. Ravi Kumar | Cardiologist   | ravi@hospital.com | 9000000001 | 120000.00 |             1 |
+-----------+----------------+----------------+-------------------+------------+-----------+---------------+
1 row in set (0.01 sec)

mysql> SELECT *
    -> FROM doctor
    -> WHERE specialization LIKE '%ologist%';
+-----------+------------------+----------------+--------------------+------------+-----------+---------------+
| doctor_id | doctor_name      | specialization | email              | phone      | salary    | department_id |
+-----------+------------------+----------------+--------------------+------------+-----------+---------------+
|         1 | Dr. Ravi Kumar   | Cardiologist   | ravi@hospital.com  | 9000000001 | 120000.00 |             1 |
|         2 | Dr. Priya Sharma | Neurologist    | priya@hospital.com | 9000000002 | 130000.00 |             2 |
+-----------+------------------+----------------+--------------------+------------+-----------+---------------+
2 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM doctor
    -> WHERE salary BETWEEN 100000 AND 130000;
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
```

5. IN Operator and LEFT JOIN

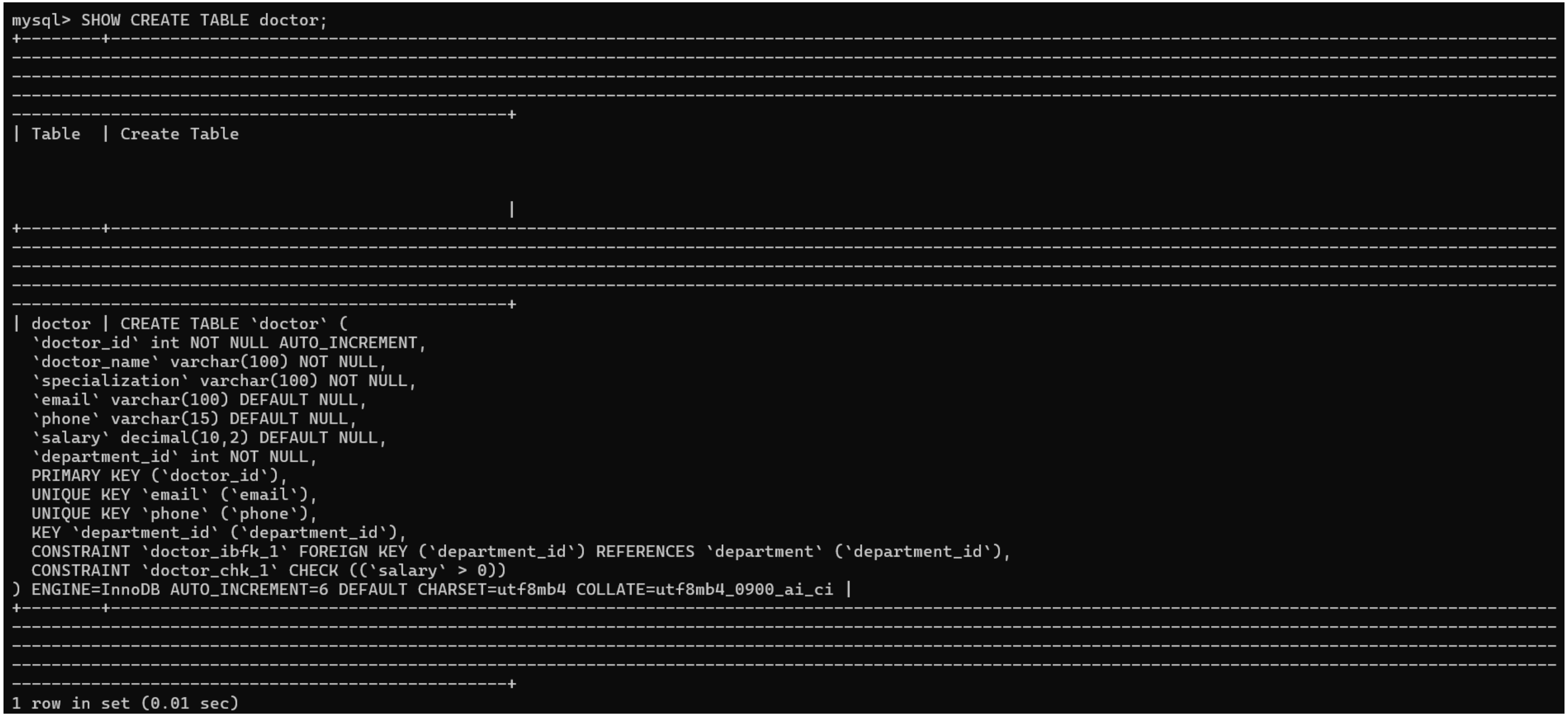
Shows filtering by multiple specializations and retrieving doctor details with matching department information.



```
mysql> SELECT *
    -> FROM doctor
    -> WHERE salary BETWEEN 100000 AND 130000;
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
| doctor_id | doctor_name      | specialization     | email              | phone      | salary    | department_id |
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
|         1 | Dr. Ravi Kumar   | Cardiologist       | ravi@hospital.com  | 9000000001 | 120000.00 |             1 |
|         2 | Dr. Priya Sharma | Neurologist        | priya@hospital.com | 9000000002 | 130000.00 |             2 |
|         3 | Dr. Arjun Rao    | Orthopedic Surgeon | arjun@hospital.com | 9000000003 | 110000.00 |             3 |
|         4 | Dr. Sneha Reddy  | Pediatrician       | sneha@hospital.com | 9000000004 | 100000.00 |             4 |
+-----------+------------------+--------------------+--------------------+------------+-----------+---------------+
4 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM doctor
    -> WHERE specialization IN
    -> ('Cardiologist', 'Neurologist');
+-----------+------------------+----------------+--------------------+------------+-----------+---------------+
| doctor_id | doctor_name      | specialization | email              | phone      | salary    | department_id |
+-----------+------------------+----------------+--------------------+------------+-----------+---------------+
|         1 | Dr. Ravi Kumar   | Cardiologist   | ravi@hospital.com  | 9000000001 | 120000.00 |             1 |
|         2 | Dr. Priya Sharma | Neurologist    | priya@hospital.com | 9000000002 | 130000.00 |             2 |
+-----------+------------------+----------------+--------------------+------------+-----------+---------------+
2 rows in set (0.00 sec)

mysql> SELECT
    ->     d.doctor_name,
    ->     d.specialization,
    ->     dept.department_name,
    ->     dept.location
    -> FROM doctor d
    -> LEFT JOIN department dept
    -> ON d.department_id = dept.department_id;
+------------------+--------------------+-----------------+----------+
| doctor_name      | specialization     | department_name | location |
+------------------+--------------------+-----------------+----------+
| Dr. Ravi Kumar   | Cardiologist       | Cardiology      | Block A  |
| Dr. Priya Sharma | Neurologist        | Neurology       | Block B  |
| Dr. Arjun Rao    | Orthopedic Surgeon | Orthopedics     | Block C  |
| Dr. Sneha Reddy  | Pediatrician       | Pediatrics      | Block D  |
```

6. Doctor Table Definition – SHOW CREATE TABLE

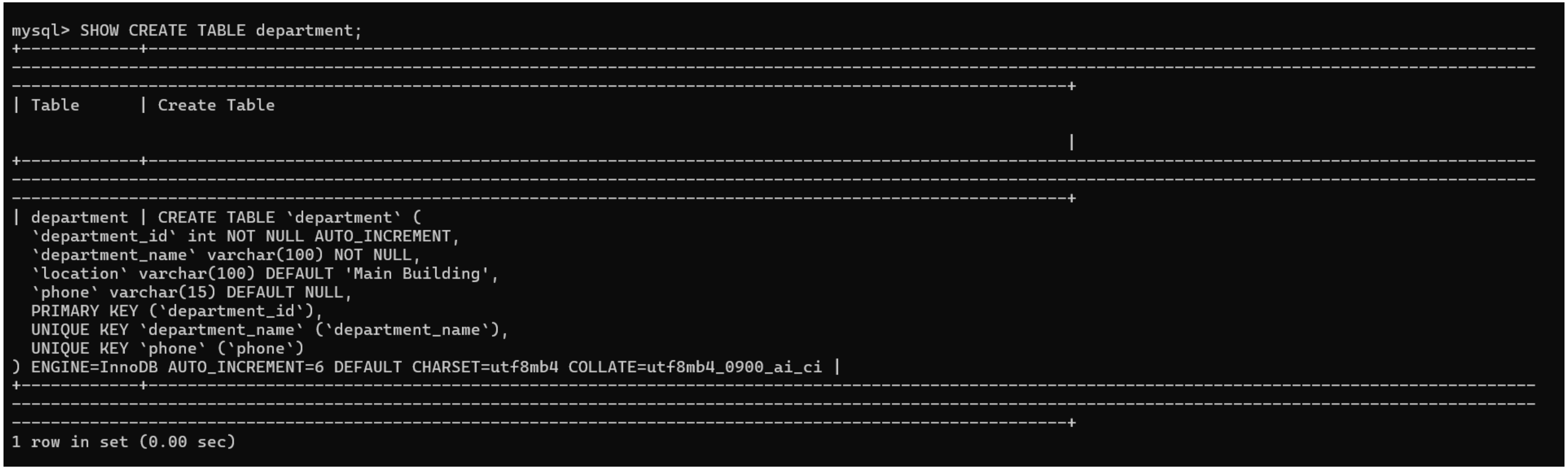
Shows the doctor table definition, including primary key, unique constraints, foreign key and salary CHECK constraint.



```
mysql> SHOW CREATE TABLE doctor;
+--------+--------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------+
| Table  | Create Table



                                                  |
+--------+--------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------+
| doctor | CREATE TABLE `doctor` (
  `doctor_id` int NOT NULL AUTO_INCREMENT,
  `doctor_name` varchar(100) NOT NULL,
  `specialization` varchar(100) NOT NULL,
  `email` varchar(100) DEFAULT NULL,
  `phone` varchar(15) DEFAULT NULL,
  `salary` decimal(10,2) DEFAULT NULL,
  `department_id` int NOT NULL,
  PRIMARY KEY (`doctor_id`),
  UNIQUE KEY `email` (`email`),
  UNIQUE KEY `phone` (`phone`),
  KEY `department_id` (`department_id`),
  CONSTRAINT `doctor_ibfk_1` FOREIGN KEY (`department_id`) REFERENCES `department` (`department_id`),
  CONSTRAINT `doctor_chk_1` CHECK ((`salary` > 0))
) ENGINE=InnoDB AUTO_INCREMENT=6 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+--------+--------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------+
1 row in set (0.01 sec)
```

7. Department Table Definition – SHOW CREATE TABLE

Shows the department table definition, including primary key, unique constraints and the default location value.



```
mysql> SHOW CREATE TABLE department;
+------------+----------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------+
| Table      | Create Table

                                                                                                            |
+------------+----------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------+
| department | CREATE TABLE `department` (
  `department_id` int NOT NULL AUTO_INCREMENT,
  `department_name` varchar(100) NOT NULL,
  `location` varchar(100) DEFAULT 'Main Building',
  `phone` varchar(15) DEFAULT NULL,
  PRIMARY KEY (`department_id`),
  UNIQUE KEY `department_name` (`department_name`),
  UNIQUE KEY `phone` (`phone`)
) ENGINE=InnoDB AUTO_INCREMENT=6 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+------------+----------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------+
1 row in set (0.00 sec)
```